

Cluster Robust Variance Estimation

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1 Abstract

This thesis studies inference in linear regression models with clustered data, with a particular focus on the consistency and interpretation of cluster-robust variance estimators (CRVE). The motivation stems from the widespread use of clustering adjustments in applied work and from recent contributions questioning their robustness when clustering is not required by the data-generating process. The first part of the thesis introduces the problem of variance estimation for the OLS estimator and clarifies why clustering matters for asymptotic inference. After defining what clustered data are, the literature review discusses the framework proposed by Abadie, Athey, Imbens, and Wooldridge, which reframes clustering as a feature of sampling and assign-

ment design rather than residual correlation, and highlights the conditions under which CRVE or White's heteroskedasticity-robust estimator is appropriate. The core of the thesis develops a general setup with heterogeneous treatment effects across clusters and decomposed error terms. Different assumptions on the number of clusters are considered, together with the role of cluster-level endogeneity. A simulation framework is then introduced to generate both exogenous and endogenous clustered data, allowing population-level exogeneity to hold while within-cluster endogeneity is present. Using Monte Carlo experiments, the thesis studies the consistency of cluster-specific estimators and of the pooled OLS estimator for the average partial effect. The results show that consistency of the average effect can be achieved even in the presence of cluster-level endogeneity, provided that population-level exogeneity holds and all clusters are observed. The final part of the thesis focuses on variance estimation. It analyzes how cluster-level endogeneity affects the asymptotic variance of the pooled estimator and shows why this effect does not average out across clusters. The CRVE matrix is derived formally, its convergence properties are discussed, and its difference from White's variance estimator is made explicit. Monte Carlo evidence illustrates the conditions under which each variance estimator is consistent and clarifies why CRVE should not be used as a default robustness check.

2 Introduction

In constructing the confidence intervals of the OLS estimator $\hat{\beta}_{ols}$, it is the asymptotic distribution of

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i$$

that determines the estimator's asymptotic distribution. Assuming that $E[X_i u_i] = 0$, the Lindernberg-Levy CLT apply:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_i \text{Var} (X_i u_i) \right)$$

But this hold true only if the individual observations are all independent, in fact summing 2 random variables X_1 and X_2 we get another random variable (call it Z) with expected value equal to $E[X_1] + E[X_2]$ and variance equal to $\text{Var} (X_1) + \text{Var} (X_2) + 2 \cdot \text{cov} (X_1, X_2)$. Summing N random variables $(X_1, \dots, X_n)$, assume $E[X_i] = 0$, we get a new random variable $Z_N = \sum_{i=1}^N X_i$:

$$Z_N \sim \left(0, \sum_{i=1}^N \text{Var} (X_i) + 2 \sum_{i \neq j}^N \sum_j^N \text{cov} (X_i, X_j) \right)$$

if all the X_i s are **independent** all the joint moments (and so also the convariancers) are equal to zero and, dividing by $\sqrt{N}$:

$$\frac{1}{\sqrt{N}} (Z_N) \sim \left(0, \frac{1}{N} \sum_{i=1}^N \text{Var} (X_i) \right)$$

So the variance of Z_N scaled by $\frac{1}{\sqrt{N}}$ is the average variance of all the X_i s. But here comes the problem, if our data are clustered we do not have that all the observation are independent, but rather we have that the observation of all the individuals inside a cluster are correlated, so we would have that

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_{i=1}^N \text{Var} (X_i u_i) + \frac{2}{N} \sum_{i \neq j}^N \sum_{j=1}^N \text{cov} (X_i u_i, X_j u_j) \right)$$

The variance estimator that is most commonly used, the White variance estimator assumes that all the covariances in the expression above are 0, and so estimates only $\frac{1}{N} \sum_{i=1}^N \text{Var} (X_i u_i)$, so it would not consistently estimate the asymptotic variance.

2.1 Why we estimate $\text{Var} (\hat{\beta})$

To follow what comes next, it is essential to clarify **what we are actually estimating** when we estimate the variance of $\hat{\beta}_{\text{ols}}$. After all, $\hat{\beta}_{\text{ols}}$ is just a number—so what does it mean to speak of its variance?

Is it a random variable? Does it have a distribution? Yes. The distribution we refer to is the distribution of $\hat{\beta}_{\text{ols}}$ **across repeated samples**, i.e., the values it would take if we changed the individuals drawn from the population. Since $\hat{\beta}_{\text{ols}}$ is a function of random variables (X_i and u_i), if we knew their full population distributions, we would know the exact value of β . But because we only observe a sample, our estimate varies from sample to sample. In this sense, $\hat{\beta}_{\text{ols}}$ is a random variable and therefore has a distribution.

2.2 What are clustered data?

Let's take a step back and ask ourselves what does it mean that our data are clustered? The defining feature of a cluster is that individuals within it exhibit some form of **unobserved** correlation.

Suppose our sample contains N individuals, each with an observed trait X . We can represent these as $X_{ig} = (X_{11}, \dots, X_{N_1 1}, X_{12}, \dots, X_{N_2 2}, \dots, X_{N_G G})$, where i indexes individuals within a cluster and g indexes the cluster.

We can think of all these X_{ig} as being drawn from the same population distribution, so they are **identically distributed**. Let us assume this distribution has mean 0.

However, individuals belonging to the same cluster are **not independent**. Their joint distribution is not simply the product of their marginals, their covariances may not be zero.

Saying that observations within a cluster are correlated means that knowing the value of one individual gives information about another individual in the same cluster. If individuals i and j belong to the same cluster g , then:

$$P(X_i > E[X] \mid X_j > E[X], i, j \in g) > P(X_i > E[X])$$

and similarly:

$$P(X_i < E[X] \mid X_j < E[X], i, j \in g) > P(X_i < E[X]).$$

outcomes inside the same cluster tend to move together relative to the overall mean—they are positively associated. If we focus on a single cluster g , we can view its observations as:

$$X_g = \{X_i \mid \text{individual } i \in \text{cluster } g\}$$

Each cluster is a subpopulation with its own distribution, and the overall population is the mixture of all G cluster distributions.

3 Literature Review

The starting point of this thesis is the framework proposed by Abadie, Athey, Imbens, and Wooldridge (A. Abadie, S. Athey, G. W. Imbens, and J. M. Wooldridge [1]), which reframes clustering adjustments as a problem of sampling and assignment design rather than residual correlation. The authors argue that the commonly used CRVE is not robust in the following sense: when clustering is not required, the CRVE does not consistently estimate the asymptotic variance of $\hat{\beta}_{OLS}$, so that using it as a precautionary default is incorrect. Moreover, they emphasize that a discrepancy between the variance estimated using White’s heteroskedasticity-robust estimator and that obtained using the CRVE does not, by itself, imply that the CRVE is the correct estimator. So what does? What the authors claim is that cluster adjustment matters not because errors or regressors are correlated within clusters, but because of how the data are generated. What drives the need for clustering adjustment is the design: either the sampling scheme, i.e. the probability of including an individual depends on the cluster she belongs to. Or the experimental design, i.e. treatment assignment itself depends on cluster membership, as in the case of region-specific policies.

In this sense, correlations of residuals or regressors inside clusters are neither necessary nor sufficient to justify cluster-robust adjustments.

They conclude that the choice between CRVE and White must depend on the estimand, not on the observed correlation. If the clusters in the sample are only a small subset of many clusters in the population, then CRVE is appropriate. If instead all clusters are observed and both treatment and sampling are i.i.d., White is correct.

CRVE tends to be conservative, except in special cases: when treatment effects are homogeneous, when the sample includes only a few clusters drawn from a much larger population, or when only a negligible fraction of units per cluster is sampled.

4 General setup

In this section, we lay out the general setup that will guide the rest of the thesis. We will assume the following model

$$y_{ig} = \beta_g \mathbf{X}_{ig} + \gamma \mathbf{Z}_g + u_{ig}$$

So there are two groups of regressors, the one that varies only at the cluster level and the one that varies also at the individual level. We assume that there are heterogeneous effect across clusters, this means that the true parameter β is different in every cluster. The average across clusters of β_g we will call β_{APE} , where APE stands for Average Partial Effect. We model u_{ig} as having a cluster component and an individual-and-cluster component:

$$u_{ig} = c_g + v_{ig}$$

Where $v_{ig} \sim (0, \sigma_v^2)$, $c_g \sim (0, \sigma_c^2)$ but of course c_g varies only at the cluster level. In this way the expected value of the error given that individual i is in cluster g is:

$$\mathbb{E}_g(u_{ig}) = c_g$$

This can be thought of as a cluster specific intercept, and as long as we include an intercept in our estimated model that varies for each cluster we can assume w.l.o.g that $\mathbb{E}_g[u_{ig}] = 0 \forall g$.

4.1 Assumptions we make

We must begin by defining precisely the assumptions that we are willing make about the data generating process.

The first choice is whether we treat the number of clusters as fixed and fully observed in every sample—for example, U.S. states ($G = 50$) when estimating the effect of a federal policy. Alternatively, we may assume an increasing number of clusters, where the specific clusters included can vary from sample to sample.

In both cases there is a two stage sampling scheme going on: in the first stage the clusters are sampled, in the second one the individuals inside the sampled clusters are drawn.

4.1.1 Fixed amount of clusters

- in the first stage all the population parameters are drawn. This include γ the vector of length G with all the β_g , and the cluster specific distributions for X_g and u_g . This stage only happens one, all the parameters from this stage are fixed.
- The second stage is a draw from the cluster distributions of (X, u) of the individual observations that will make our sample. This stage re-happens every time we take a sample from the population, this is the (only) source of randomness that there is in this setup.

4.1.2 Infinite amount of clusters

If the amount of clusters that we possibly have is not finite the first sampling stage reoccurs every time we sample. So there are two sources of randomness:

1. which clusters we draw from the possibly infinite population of clusters and,
2. which individuals we draw from those clusters.

4.2 Endogeneity

Then we must decide whether the sample we are working with presents cluster-level endogeneity, i.e. if

$$\mathbb{E}[X_{ig}c_g \mid i \in g] \neq 0.$$

In order for $\hat{\beta}_{ols}$ to be consistent we need exogeneity, meaning that regressors must be uncorrelated with the errors. However, this condition may hold on average (across clusters) while failing within individual clusters: some clusters may display correlation between the cluster-specific component of the error and the regressors, as discussed by Y. Mundlak [2]. When this happens, we

may still assume that these correlations can cancel out across clusters, so that exogeneity holds only in the aggregate and a pooled OLS estimator is consistent.

5 Simulating a dataset

We will simulate data in two ways, in the first one we will make so that the expected value of the score is 0 in all clusters, in the second one the expected value of the score will be 0 only at the population level.

First we draw a random vector β_g from a normal distribution with mean β_{APE} and variance σ_β^2 .

```
include("../src/ClusterSimulation.jl")
G = 100 #Number of clusters
Ng = 500 # Number of individuals per cluster

β = rand(Normal(2,4),G)
```

```
100-element Vector{Float64}:
 5.167836049127602
-1.5971549516224872
-1.1052817125857897
-1.804261339887319
 5.623442492002585
 7.443169150942235
 8.2437210679444
 9.102947997997267
 1.0088501746308012
 0.5582077716520422
-3.751119855090284
 0.8778010922474127
-0.30361439654164935
 ⋮
 3.802377584900829
13.567252697917786
-2.9558247510934086
 6.456504006548233
 4.738817446136096
 8.107879180677376
 3.195843350610888
 3.2737713697697126
 2.329694695426956
-0.838906296551233
-2.9949425590609042
-3.405230333389821
```

We then simulate our dataset in two steps, according to the following model:

$$y_{ig} = \beta_g x_{ig} + (c_g + v_{ig})$$

In the first step we simulate the population parameters, and in the second we draw from them.

5.1 Description of the Two Steps

```
pop = hePopulation(G, β, randn(G))
```

Population with 100 clusters and with heterogeneous effect

This line creates a heterogeneous-effects population.

- **G** is the number of clusters.

- **β** is the vector of betas.

- **randn(G)** produces the vector $c_g \sim \mathcal{N}(0, 1)$.

The constructor returns a population struct that stores the cluster-level parameters used to simulate data.

```
sam = sample(pop, Ng)
```

Cluster sample struct

The line above generates a sample from the population.

- **Ng** is the number of observations per cluster.

The function draws data for all **G** clusters, each containing **Ng** observations, and returns a sample struct built according to the population's parameters.

I can then estimate $\hat{\beta}_g$ in each cluster and then broadcast it to all clusters in our sample, getting **G** estimates.

Each of these

5.2 Endogenous clusters

The second way in which we'll simulate our data is as follow.

We want to simulate data such that:

$$\mathbb{E}[X_{ig}u_{ig} \mid i \in \text{cluster } g] = \mu_g$$

for each cluster g , and

$$\mathbb{E}[X_{ig}u_{ig}] = \mathbb{E}[\mathbb{E}[X_{ig}u_{ig} \mid i \in \text{cluster } g]] = \mathbb{E}[\mu_g] = 0$$

Across all clusters, the overall expectation is zero because μ_g has mean zero. We first draw from a normal distribution $\mathcal{N}(0, \sigma_\mu^2)$ a vector μ of length G , with all the cluster's mean μ_g . Note that in doing so we are taking a sample of size G from a theoretically infinite population of clusters. Then we:

1. Draw two i.i.d. standard normal variables:

$$Z_1 \sim \mathcal{N}(0, 1), \quad Z_2 \sim \mathcal{N}(0, 1)$$

Z_1 and Z_2 are zero-mean and independent. The covariance between X and u will come entirely from the shared Z_1 component.

2. Construct within each cluster g :

$$X_g = s_g Z_1$$

$$u_g = a_g Z_1 + b_g Z_2$$

a_g controls the strength of the correlation within the cluster, b_g let u have positive variance.

$$a_g = \frac{\mu_g}{s_g}, \quad b_g = \sqrt{s_g^2 - a_g^2}.$$

To make sure that b_g exist and is positive we need $s_g^4 \geq \mu_g^2$. So we choose:

$$s_g^2 = |\mu_g| + v_0 \quad \text{with } v_0 > 0$$

So that:

$$\mathbb{E}[X | g] = \mathbb{E}[u | g] = 0$$

and

$$\mathbb{E}[Xu | g] = \text{Cov}(X, u | g) = s_g a_g = \mu_g.$$

We then compute Y_{ig} choosing a value for β .

$$Y_{ig} = X_{ig}\beta + u_{ig}$$

Let us now check that the expected value of the score is 0 in the whole population but not in each clusters.

```
beta2 = 2
μg = rand(Normal(0,4),500)
endoPop = clPopulation(500, μg, beta2)
endoSam = sample(endoPop,100)
S = checkScore(endoSam)
```

```
500-element Vector{Float64}:
 1.4229153101257759
 0.9647669048266688
-5.83144138446916
 0.3311324206024259
 1.0627326663587668
 3.3325197850194037
-0.6969703260252406
-2.094319787688606
-1.9439750419110546
 3.577651126207277
```



```

-2.5401007951972616
-0.542629803186743
 1.990042088338094
  ⋮
 1.3151839491844546
-1.3573560291385005
-0.04091178074365481
 5.237225803315398
-3.7003890581956607
-1.9724564300213914
 4.680087815154678
 0.7714612088295856
-2.360861933158198
 1.1676558651477713
-3.0903004411859856
 1.3090955385042964

```

The above is the score in each cluster, now let's check the mean and variance.

```
mean(S), var(S)
```

```
(-0.017896486987056332, 4.699338720637391)
```

We can see that the mean is close to 0 and the variance is close to the expected variance of μ_g .

6 Consistency

We can discuss two ways in which our estimator can be consistent, the first one is that:

$$\widehat{\beta}_g \xrightarrow{p} \beta_g$$

meaning that each of the G estimated parameters is consistent for its counterpart in the population. This would require that the expected value of the score is 0 in all clusters, i.e. no cluster-level endogeneity. The second way in which our estimator may be consistent is that:

$$\hat{\beta}_{\text{pols}} \xrightarrow{p} \beta_{\text{APE}}$$

For this to be the case we only need population exogeneity.

6.1 Montecarlo without endogeneity

We will now use a Montecarlo method to test whether we can get a consistent estimate of the G parameters if there is no cluster endogeneity. We will estimate the following model:

$$\mathbf{y} = \mathbf{X}_g \hat{\beta}_{Ng} + \mathbf{u}_g$$

Where $\hat{\beta}_g = \begin{bmatrix} \alpha_g \\ \beta_g \end{bmatrix}$. Adding a cluster-specific intercept lets the model absorb cluster-level fixed effects in the residuals.

```
histogram(
  permutedims(stack(bols.(sam.allclusters)))[:,2] - β,
  bins = 15
)
```

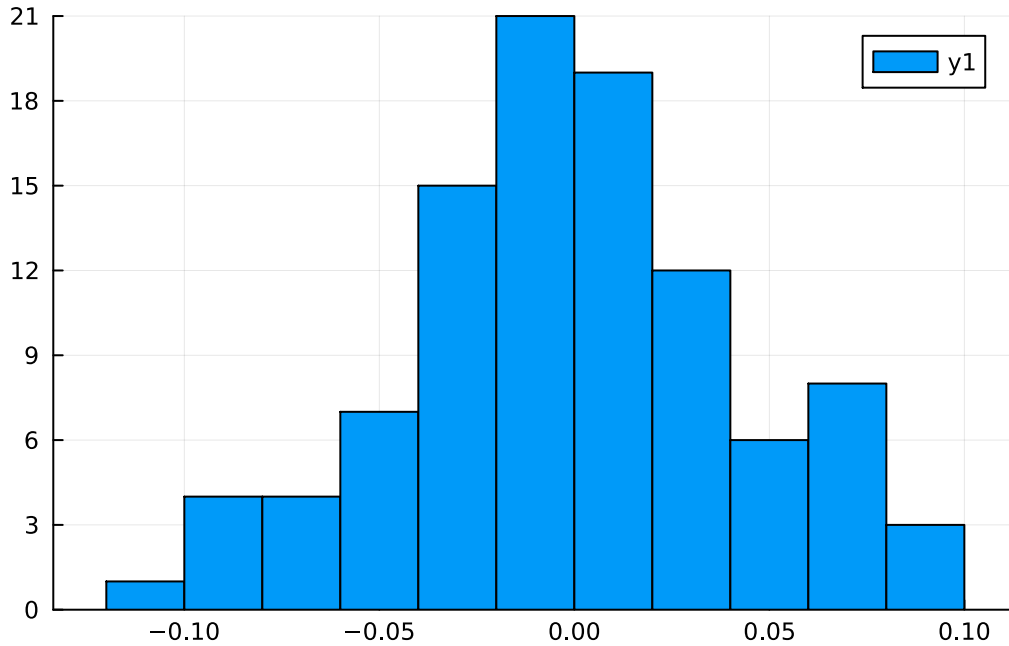


Figure 1: Histogram of the difference between the estimated parameters and the true one in all clusters

In Figure 1 we can see a plot of the difference between the estimated value for β_g and the true value, it is clear from the graph that all the estimated parameters are consistent for their population counterpart. Now we'll run a Montecarlo simulation, with 100 iterations, we will simulate for each iteration 50 clusters of 100 people and compute $\hat{\beta}_g$ for each cluster. We will then compute the mean across simulation of the difference between the estimated β_g and the true parameter.

```
betahatDiff = montecarloClusterWise(pop,100,100) .- β
histogram(mean(betahatDiff, dims = 2), bins = 15)
```

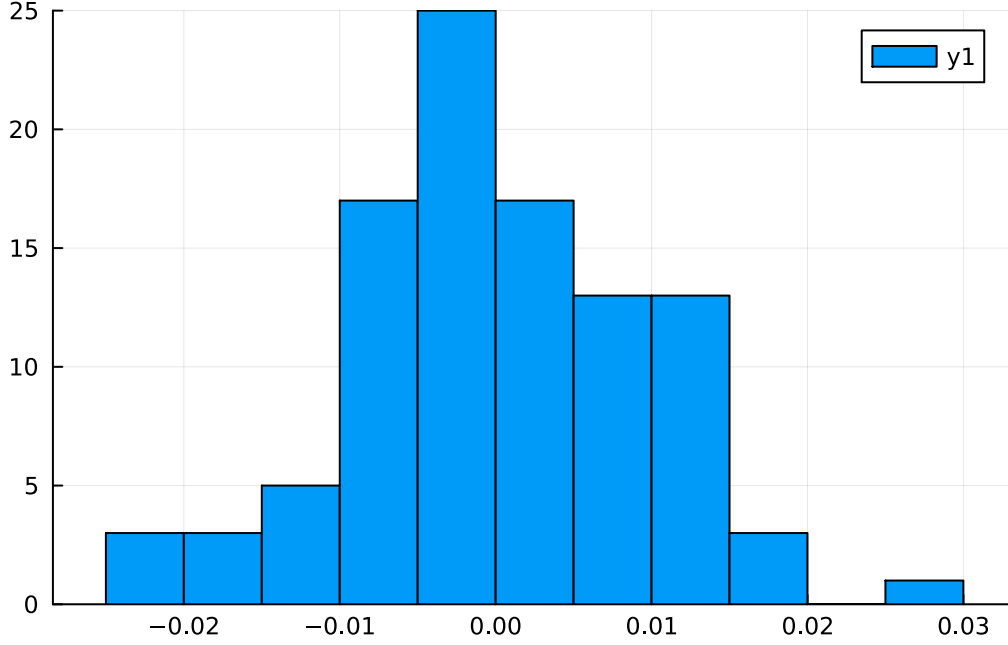


Figure 2: Histogram of the mean difference between estimated and true parameters across simulation in the montecarlo experiment

As can be clearly seen from Figure 2 there is consistency across all clusters.

6.2 APE

Since β_{APE} is simply the average of all the β_g , if each β_g can be consistently estimated, then β_{APE} can also be consistently estimated. But what happens if there is cluster-level endogeneity?

In that case, consistency is still achievable—as long as endogeneity does not arise at the population level. The key point is that estimating $\hat{\beta}_{\text{pols}}$ is equivalent to averaging the $\hat{\beta}_g$ across clusters. For each cluster we have

$$\hat{\beta}_g \xrightarrow{p} \beta_g + \frac{\sum_{i \in g} S_i}{\sum_{i \in g} \mathbf{x}_i' \mathbf{x}_i}$$

where $S_i = X_i u_i$ is the score. Averaging this over all G clusters gives:

$$\hat{\beta}_{\text{pols}} = \frac{1}{G} \sum_g \hat{\beta}_g \xrightarrow{p} \beta_{\text{APE}} + \frac{\sum_g \sum_{i \in g} S_i}{\sum_g \sum_{i \in g} \mathbf{x}_i' \mathbf{x}_i}$$

As long as we observe all clusters, the numerator of the second term is equal to 0, and the estimator is therefore consistent.

6.3 Consistency as $N_g \xrightarrow{p} \infty$

We want to test whether our estimate of the average partial effect gets better if the number of people that we draw in each cluster increases. In order to do so we ran a series of monte-carlo simulation increasing N_g .

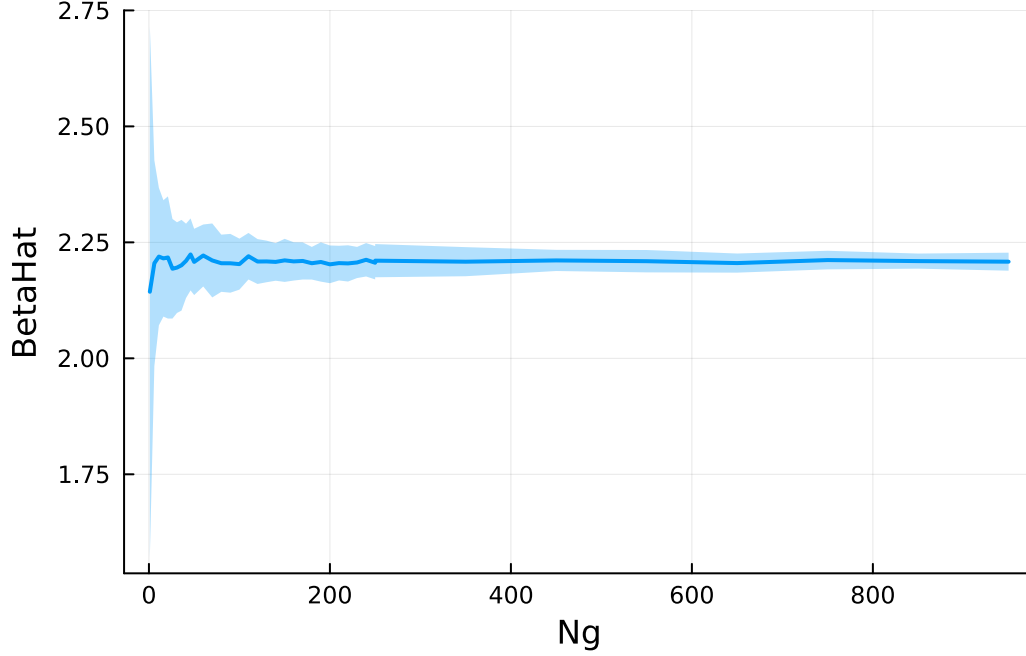


Figure 3: Consistency of betahat as N_g

What we see in Figure 3 is the result of the monte-carlo simulation, the line is the average estimates for each N_g , while the shaded area is the standard deviation, we can clearly see from the graph that as the number of observations inside a cluster grows $\hat{\beta}_{ols} \xrightarrow{p} \beta_{APE}$.

7 Variance Estimation

In each cluster we have that

$$\hat{\beta}_g \xrightarrow{p} \beta_g + \frac{\sum_{i \in g} S_i}{\sum_{i \in g} \mathbf{x}_i' \mathbf{x}_i}$$

Where S_i is the score $X_i u_i$.

$$(\hat{\beta}_g - \beta_g) \xrightarrow{d} \mathcal{N}(0, V_g)$$

$$V_g = \sum_{i \in g} \mathbf{x}_i' \mathbf{x}_i \left(\text{Var} \left(\sum_{i \in g} S_i \right) \right) \sum_{i \in g} \mathbf{x}_i' \mathbf{x}_i$$

Since inside each cluster observations can be considered i.i.d.

$$\text{Var} \left(\sum_{i \in g} S_i \right) = \sum_{i \in g} \text{Var} (S_i)$$

The variance is then

$$\mathbb{E}(S_{ig}^2) - [\mathbb{E}(S_{ig})^2]$$

Now if we want to estimate β_{APE} we note that it is equal to the average across G of the various β_g , and since the various $\hat{\beta}_g$ are independent, the variance of their mean will be the mean of their variances.

$$\sum_g (\hat{\beta}_g - \beta_g) = \hat{\beta}_{\text{pols}} - \beta_{\text{APE}} \stackrel{a}{\sim} \mathcal{N} \left(0, \frac{1}{G} \sum_g \text{Var} (\hat{\beta}_g) \right)$$

As noted before if we have enough clusters:

$$\sum_g \sum_{i \in g} S_i \xrightarrow{p} \mathbb{E}_g(S_{ig}) = 0$$

So we get a consistent estimate even if there is cluster level endogeneity. But this not apply to the variance. That is because

$$\sum_g \sum_{i \in g} S_i^2 \xrightarrow{p} \mathbb{E}_g(S_{ig}^2) \neq 0$$

So that there is not averaging out across clusters that make the second term disappear. This means that the asymptotic variance of $\hat{\beta}_{\text{pols}}$ will be different depending on whether there is cluster-level endogeneity or not, and the difference will be equals to:

$$V_{\text{endo}} - V_{\text{exo}} = \frac{\mathbb{E}_g(\mu_g^2)}{\sum_i \mathbf{x}_i' \mathbf{x}_i}$$

7.1 All cluster observed and no endogeneity

So if there is no endogeneity i can consistently estimate the AVar of each $\hat{\beta}_g$ using White variance estimator, let's test this using the montecarlo method.

```
meanbeta, sdbeta = montecarlo(pop,1000,Ng)
D = Normal(meanbeta,sdbeta)

x = range(meanbeta - 4*sdbeta,meanbeta + 4*sdbeta; length = 1000)
j = pdf.(D,x)

plot(x,j, linewidth=2, label="Montecarlo")

betahat = bols(sam)[2]
sigma = sqrt(whiteAvar(sam)[2,2])
```

```

d = Normal(betahat, sigma)

y = pdf.(d, x)

plot!(x, y, linewidth= 2, label = "White")

```

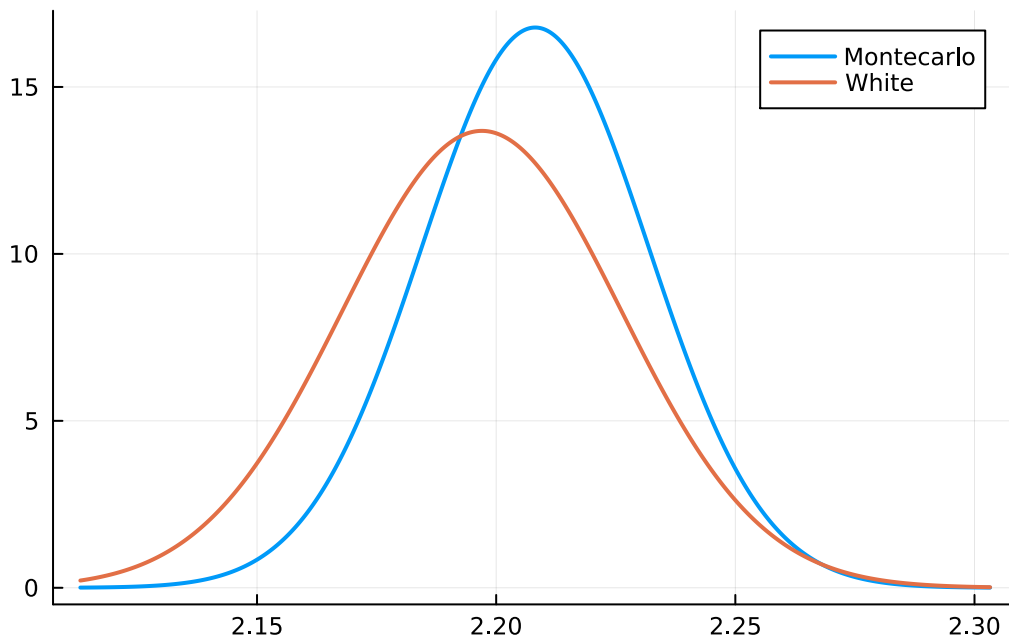


Figure 4: Consistency of White Variance estimator without cluster level endogeneity

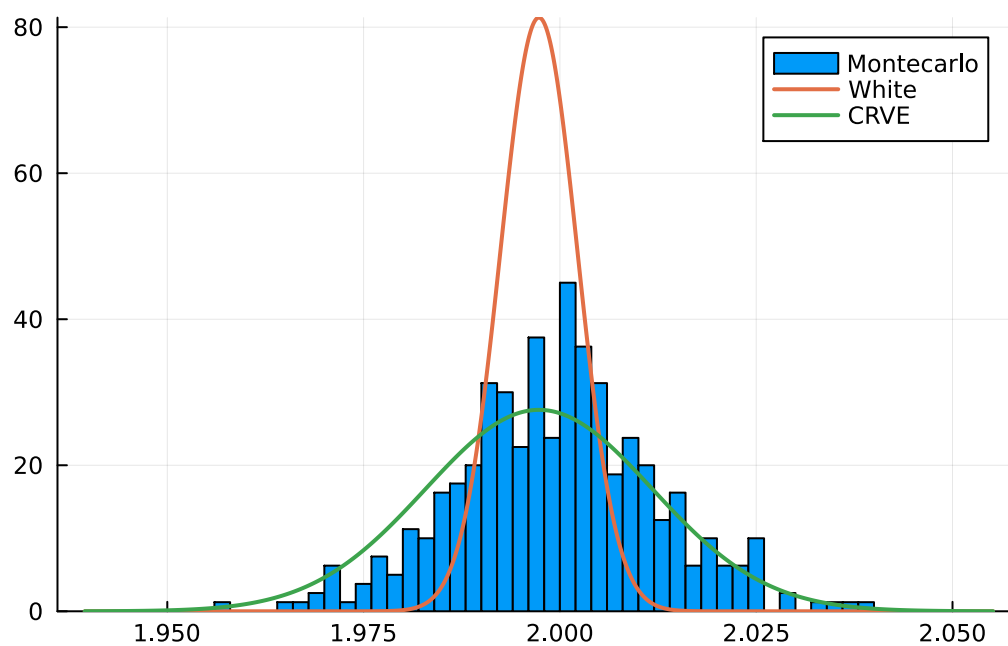
Figure 4

```

betahat = bols(endoSam)[2]
white = sqrt(whiteAvar(endoSam)[2,2])
crve = sqrt(CRVE(endoSam)[2,2])
histogram(montecarlo(endoPop,400,Ng,true,true), bins=50,normalize=:pdf,
label="Montecarlo")
d = Normal(betahat, white)
D = Normal(betahat, crve)

x = range(betahat - 4 * crve, betahat + 4 * crve; length = 1000)
y = pdf.(d,x)
Y = pdf.(D,x)
plot!(x,y, linewidth =2, label = "White")
plot!(x,Y, linewidth =2, label = "CRVE")

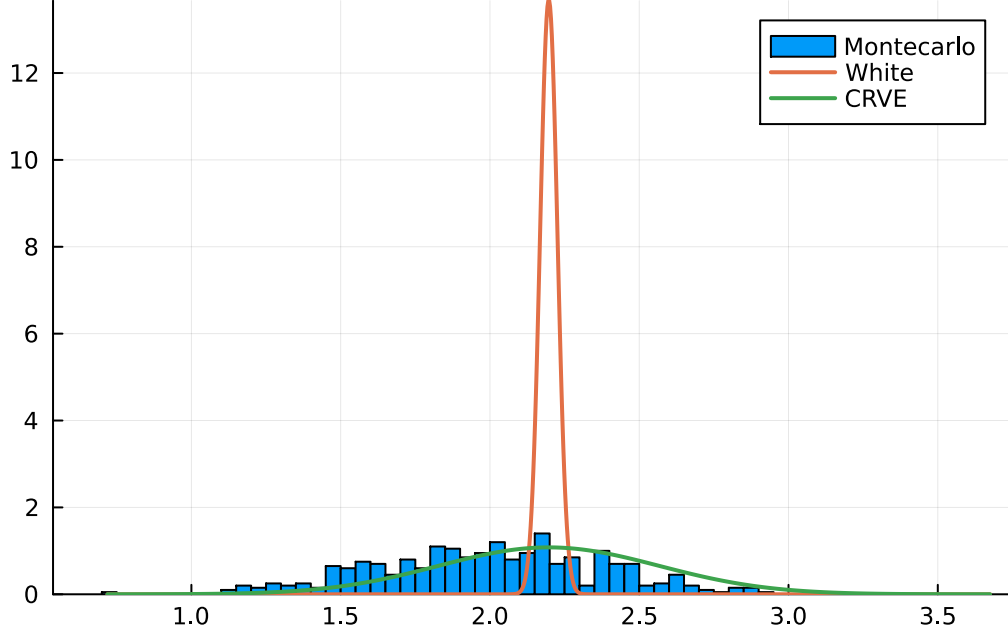
```



```

betahat = bols(sam)[2]
white = sqrt(whiteAvar(sam)[2,2])
crve = sqrt(CRVE(sam)[2,2])
histogram(montecarlo(pop,400,Ng,true,true),bins=50,normalize=:pdf,
label="Montecarlo")
d = Normal(betahat, white)
D = Normal(betahat, crve)
x = range(betahat - 4 * crve, betahat + 4 * crve; length = 1000)
y = pdf.(d,x)
Y = pdf.(D,x)
plot!(x,y, linewidth =2, label = "White")
plot!(x,Y, linewidth =2, label = "CRVE")

```



8 CRVE Matrix

8.1 Notation

We use the same notation as in [3]:

$$\mathbf{X}_{ig} = [x_{1ig}, x_{2ig}, \dots, x_{kig}]$$

$$\mathbf{X}_g = \begin{bmatrix} \mathbf{X}_{ig} \\ \mathbf{X}_{jg} \\ \dots \\ \mathbf{X}_{N_g g} \end{bmatrix} = \begin{bmatrix} x_{1ig}, & x_{2ig}, & \dots & x_{kig} \\ x_{1ij}, & x_{2jg}, & \dots & x_{kjg} \\ \dots & & & \\ x_{1N_g g}, & x_{2N_g g}, & \dots & x_{kN_g g} \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_{ig} \\ \mathbf{X}_{jg} \\ \dots \\ \mathbf{X}_{N_g g} \\ \mathbf{X}_{ih} \\ \mathbf{X}_{jh} \\ \dots \\ \mathbf{X}_{N_h h} \\ \dots \\ \mathbf{X}_{iG} \\ \mathbf{X}_{jG} \\ \dots \\ \mathbf{X}_{N_G G} \end{bmatrix}$$

—

The idea behind CRVE is that clusters are independent one from the other. Since inference requires independence across a big number of units, we base it on the number of clusters rather than the number of individuals. Assuming that $\mathbb{E}[\mathbf{X}'_g \mathbf{u}_g] = 0$, $\hat{\beta}_{ols}$ will be a consistent estimator of the population parameter β if $G \rightarrow \infty$ [4]. To understand this note that computing β :

$$(X'X)^{-1}(X'y)$$

is the same as doing:

$$\hat{\beta}_{ols} = \left(\sum_g^G \mathbf{X}_g' \mathbf{X}_g \right)^{-1} \left(\sum_g^G \mathbf{X}_g' y_g \right)$$

so

$$(\hat{\beta}_{ols} - \beta) = \left(\frac{1}{G} \sum_g^G \mathbf{X}_g' \mathbf{X}_g \right)^{-1} \left(\frac{1}{G} \sum_g^G \mathbf{X}_g' \mathbf{u}_g \right)$$

and applying the Weak Law of Large Numbers J. M. Wooldridge [4]

$$\frac{1}{G} \sum_g^G \mathbf{X}_g' \mathbf{u}_g \xrightarrow{p} \mathbb{E}[\mathbf{X}'_g \mathbf{u}_g] = 0$$

so $\hat{\beta}_{ols}$ will be consistent. Now let us derive it's asymptotic distribution:

$$\sqrt{G}(\hat{\beta}_{ols} - \beta) = \left(\frac{1}{G} \sum_g^G \mathbf{X}_g' \mathbf{X}_g \right)^{-1} \left(\frac{1}{\sqrt{G}} \sum_g^G \mathbf{X}_g' \mathbf{u}_g \right)$$

The first term is by assumption $A^{-1} + o_p(1)$, while the second is $\overset{a}{\sim} \mathcal{N}(0, \mathbb{E}[\mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g])$ So

$$\sqrt{G}(\hat{\beta}_{\text{ols}} - \beta) \xrightarrow{d} \mathcal{N}(0, A^{-1} \mathbb{E}[\mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g] A^{-1})$$

8.2 Formula

$$\hat{V}_{\text{clu}} \hat{\beta} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{B} (\mathbf{X}' \mathbf{X})^{-1} = (\mathbf{X}' \mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{X}_g' \hat{\mathbf{u}}_g \hat{\mathbf{u}}_g' \mathbf{X}_g \right) (\mathbf{X}' \mathbf{X})^{-1}$$

Define:

$$\mathbf{s}_g \equiv \mathbf{X}_g' \hat{\mathbf{u}}_g = \begin{bmatrix} X_{11g} & X_{12g} & \cdots & X_{1N_gg} \\ X_{21g} & X_{22g} & \cdots & X_{2N_gg} \\ \vdots & \vdots & \ddots & \vdots \\ X_{k1g} & X_{k2g} & \cdots & X_{kN_gg} \end{bmatrix} \begin{bmatrix} \hat{u}_{1g} \\ \hat{u}_{2g} \\ \vdots \\ \hat{u}_{N_gg} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{N_g} X_{1ig} \hat{u}_{ig} \\ \sum_{i=1}^{N_g} X_{2ig} \hat{u}_{ig} \\ \vdots \\ \sum_{i=1}^{N_g} X_{kig} \hat{u}_{ig} \end{bmatrix}$$

Then:

$$\mathbf{s}_g \mathbf{s}_g' = (X_g' u_g)(X_g' u_g)' = X_g' u_g u_g' X_g$$

$$\mathbf{s}_g \mathbf{s}_g' = \begin{bmatrix} \sum_{i \in g} \sum_{j \in g} X_{1ig} X_{1jg} \hat{u}_{ig} \hat{u}_{jg} & \sum_{i \in g} \sum_{j \in g} X_{1ig} X_{2jg} \hat{u}_{ig} \hat{u}_{jg} & \cdots & \sum_{i \in g} \sum_{j \in g} X_{1ig} X_{kjg} \hat{u}_{ig} \hat{u}_{jg} \\ \sum_{i \in g} \sum_{j \in g} X_{2ig} X_{1jg} \hat{u}_{ig} \hat{u}_{jg} & \sum_{i \in g} \sum_{j \in g} X_{2ig} X_{2jg} \hat{u}_{ig} \hat{u}_{jg} & \cdots & \sum_{i \in g} \sum_{j \in g} X_{2ig} X_{kjg} \hat{u}_{ig} \hat{u}_{jg} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i \in g} \sum_{j \in g} X_{kig} X_{1jg} \hat{u}_{ig} \hat{u}_{jg} & \sum_{i \in g} \sum_{j \in g} X_{kig} X_{2jg} \hat{u}_{ig} \hat{u}_{jg} & \cdots & \sum_{i \in g} \sum_{j \in g} X_{kig} X_{kjg} \hat{u}_{ig} \hat{u}_{jg} \end{bmatrix}$$

Then we sum this over all G clusters: $\mathbf{s} \mathbf{s}' = \mathbf{s} \mathbf{s}'$

So what CRVE does is to compute for each cluster the outer product of $\hat{\mathbf{S}}_g = \mathbf{X}_g' \hat{\mathbf{u}}_g$ with itself, and then average those matrices across clusters.

After pre and post-multiplication by $(\frac{1}{N} \sum_i \mathbf{x}_i \mathbf{x}_i')^{-1}$, this converges to the **asymptotic variance-covariance matrix** of the OLS estimator $\hat{\beta}_{\text{ols}}$.

8.3 Convergence

We want to show that

$$\frac{1}{G} \sum_g \mathbf{X}_g' \hat{\mathbf{u}}_g \hat{\mathbf{u}}_g' \mathbf{X}_g \xrightarrow{p} \mathbb{E}[\mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g]$$

First let's substitute for $\hat{\mathbf{u}}_g$:

$$\frac{1}{G} \sum_g \mathbf{X}_g' (\mathbf{y}_g - \mathbf{X}_g \hat{\beta}) (\mathbf{y}_g - \mathbf{X}_g \hat{\beta})' \mathbf{X}_g$$

$$\frac{1}{G} \sum_g \mathbf{X}_g' (\mathbf{X}_g \beta + \mathbf{u}_g - \mathbf{X}_g \hat{\beta}_g) (\mathbf{X}_g \beta + \mathbf{u}_g - \mathbf{X}_g \hat{\beta}_g)' \mathbf{X}_g$$

We then factor terms in order to isolate $(\beta - \hat{\beta})$

$$\frac{1}{G} \sum_g \mathbf{X}_g' [\mathbf{X}_g (\beta - \hat{\beta}_g) + \mathbf{u}_g] [\mathbf{X}_g (\beta - \hat{\beta}_g) + \mathbf{u}_g]' \mathbf{X}_g$$

As $\hat{\beta} \xrightarrow{p} \beta$, the extra terms vanish, leaving only $\mathbf{u}_g$.

$$\xrightarrow{p} \frac{1}{G} \sum_g \mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g \xrightarrow{p} [\mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g]$$

8.4 Difference from White

Let's look at the difference between the White covariance matrix and the CRVE. Imagine we have a cluster of 2 observations, the White inner matrix would look like this:

$$B_W = \begin{bmatrix} \hat{u}_1^2 & 0 \\ 0 & \hat{u}_2^2 \end{bmatrix}$$

While the CRVE would look like this:

$$B_C = \begin{bmatrix} \hat{u}_1^2 & \hat{u}_1 \hat{u}_2 & \hat{u}_1 \hat{u}_3 \\ \hat{u}_1 \hat{u}_2 & \hat{u}_2^2 & \hat{u}_2 \hat{u}_3 \\ \hat{u}_1 \hat{u}_3 & \hat{u}_2 \hat{u}_3 & \hat{u}_3^2 \end{bmatrix}$$

So when I then pre and post multiply by X :

$$X' B_W X = x_1^2 u_1^2 + x_2^2 u_2^2 = \sum_i x_i^2 u_i^2$$

$$X' B_C X = x_1^2 \hat{u}_1^2 + x_2^2 \hat{u}_2^2 + x_3^2 \hat{u}_3^2 + 2(x_1 x_2 \hat{u}_1 \hat{u}_2 + x_1 x_3 \hat{u}_1 \hat{u}_3 + x_2 x_3 \hat{u}_2 \hat{u}_3) = X' B_W X + 2 \sum_i \sum_{j \neq i} x_i x_j \hat{u}_i \hat{u}_j$$

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